AI-Based Early Parkinson's Disease Detection Using ResNet50 Transfer Learning and Explainable Artificial Intelligence

Department of Medical Engineering | Deep Learning Techniques Project | Academic Year 2026

***Abstract*—**Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting millions globally, yet its early clinical detection remains unreliable due to the subtlety of initial motor symptoms and the subjective nature of conventional diagnostic protocols. This paper presents an end-to-end AI-based diagnostic system that classifies spiral handwriting images as Healthy or Parkinson's by leveraging ResNet50 transfer learning with ImageNet-pretrained weights, supplemented by Gradient-weighted Class Activation Mapping (Grad-CAM) for prediction explainability. The classification model appends a task-specific head—Flatten, Dropout (0.5), Dense (128, ReLU), Dropout (0.3), Dense (1, Sigmoid)—atop the frozen ResNet50 backbone. Training was conducted on 72 spiral handwriting images for five epochs using the Adam optimiser at learning rate 0.0001 with binary cross-entropy loss. On a held-out test set of 30 balanced images the system achieved 80% accuracy, macro-averaged precision 0.81, recall 0.80, and F1-score 0.80. Grad-CAM heatmaps from conv5_block3_out provide spatial interpretability highlighting discriminative image regions. The system is deployed as an interactive Flask web application supporting image upload, real-time inference, confidence score visualisation, Grad-CAM overlay, and prediction logging.

Keywords*—*Parkinson's Disease, Transfer Learning, ResNet50, Grad-CAM, Explainable AI, Spiral Handwriting, Flask Deployment, Medical Image Classification, Deep Learning.

I. INTRODUCTION

Parkinson's Disease is the second most prevalent neurodegenerative disorder worldwide, surpassed only by Alzheimer's Disease in global incidence [1]. It arises from the progressive degeneration of dopaminergic neurons in the substantia nigra, leading to reduced striatal dopamine and the cardinal motor features: resting tremor, bradykinesia, muscular rigidity, and postural instability. Non-motor manifestations—micrographia, cognitive impairment, autonomic dysfunction—compound disease burden. Epidemiological projections estimate over 12 million individuals will be affected by 2040 [2].

Despite its clinical significance, PD diagnosis relies primarily on neurological examination—observer-dependent and identifying disease only after substantial dopaminergic loss has occurred. DaTscan imaging requires radioactive tracers and restricted availability, precluding routine screening. The UPDRS scoring instrument remains susceptible to inter-rater variability. These limitations motivate automated, objective, and scalable diagnostic tools.

Handwriting analysis is a promising non-invasive biomarker. Micrographia and irregularities in spiral drawing execution are well-documented preclinical indicators of Parkinson's motor dysfunction [3]. Handwriting tasks are digitally capturable at near-zero cost and without specialised clinical infrastructure, offering a compelling basis for AI-driven screening pipelines. The application of deep learning in clinical domains introduces an interpretability challenge. Regulatory frameworks demand that AI predictions be accompanied by human-understandable justifications. This paper contributes: (i) a ResNet50 transfer learning model for binary PD classification on spiral handwriting; (ii) a Grad-CAM explainability module from conv5_block3_out; and (iii) a Flask-based web deployment integrating inference, visualisation, and prediction logging.

II. LITERATURE REVIEW

Shyamala et al. [4] proposed a deep fusion architecture combining multiple feature extraction streams on HandPD, NewHandPD, and PaHaW corpora. Classification accuracy improved over single-stream architectures via fusion of convolutional and sequential feature pathways, though generalisability was constrained by limited dataset diversity and absence of independent clinical validation.

Razaq et al. [5] constructed a CNN-based framework for early-stage PD screening from handwriting and spiral images. Automated feature extraction without predefined descriptors allowed task-relevant representation discovery; however, image quality sensitivity and absence of multi-centre testing limit deployment confidence.

Sharma et al. [6] incorporated attention mechanisms into a CNN applied to spiral handwriting, paired with Grad-CAM to improve transparency. Attention modules selectively weighted relevant spatial regions and Grad-CAM confirmed clinically meaningful focus, but added computational overhead increases inference latency.

Khan et al. [7] developed a non-invasive PD screening framework grounded in spiral drawing analysis. Competitive accuracy was achieved; real-world deployment was limited and larger, more diverse datasets were identified as necessary for clinical integration.

El-Mohsen et al. [8] proposed a hybrid framework consolidating convolutional, recurrent, and attention components applied to multi-modal PD data. Multi-modal combination improved performance, but substantially increased computational complexity poses challenges in resource-constrained clinical environments.

VIII. GRAD-CAM EXPLAINABILITY MODULE

Gradient-weighted Class Activation Mapping (Grad-CAM) [10] generates a spatially localised importance map highlighting which input image regions most strongly drive a classification decision, without modifying the trained network. The gradient of the target

III. RESEARCH GAP

Synthesising the reviewed literature reveals four recurring limitations: (i) absence of prediction explainability in most frameworks; (ii) underexplored use of ResNet50 with a streamlined binary head for handwriting-based PD detection; (iii) no end-to-end clinician-accessible deployment; and (iv) lack of structured prediction logging for audit compliance. The proposed framework directly addresses all four gaps.

IV. MATERIALS AND METHODS

The proposed system follows a six-stage pipeline: (1) dataset acquisition, (2) image preprocessing, (3) ResNet50 model construction and training, (4) Grad-CAM explainability, (5) performance evaluation, and (6) Flask deployment. Figure 1 illustrates the complete system workflow from spiral image input to web dashboard output.

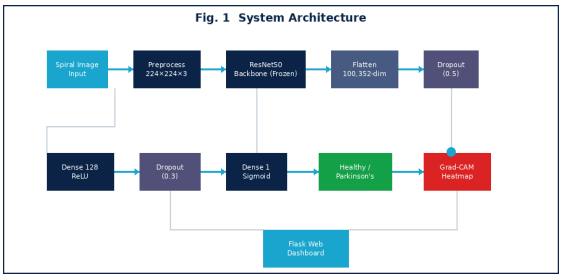


Fig. 1. End-to-end system architecture from spiral image upload through ResNet50 classification and Grad-CAM to Flask output.

V. DATASET DESCRIPTION

Experiments used the Parkinson's Spiral Handwriting Dataset in which participants drew spiral patterns under standardised conditions—clinically sensitive to upper-limb tremor. Two classes: Healthy (control) and Parkinson's (confirmed PD). Partitioned into 72 training and 30 test images (~70:30). The balanced test set (15 Healthy, 15 Parkinson's) ensures accuracy is meaningful. Table I summarises composition.

TABLE I

Dataset Composition and Class Distribution

Subset	Healthy	Parkinson's
Training	36	36
Testing	15	15
Total	**51**	**51**

VI. IMAGE PREPROCESSING

All images were resized to 224×224×3 pixels using bilinear interpolation. Pixel intensities were normalised via tf.keras.applications.resnet50.preprocess_input(), applying channel-wise mean subtraction corresponding to ImageNet statistics (BGR means: [103.939, 116.779, 123.680]). At inference time within Flask, the identical sequence PIL.Image.load_img() → img_to_array() → expand_dims() → preprocess_input() preserves pipeline consistency between training and deployment, preventing accuracy degradation from preprocessing domain shift.

VII. RESNET50 TRANSFER LEARNING ARCHITECTURE

ResNet50 [9] is a 50-layer deep residual network pre-trained on ImageNet (1.2M images, 1,000 classes). Its defining innovation is the residual connection routing block inputs directly to outputs via identity shortcuts, reformulating learning from H(x) to F(x)=H(x)−x and stabilising gradient flow in deep networks:

$$y = F(x, \{W_i\}) + x \quad (1)$$

In the proposed system the ResNet50 backbone is fully frozen—preventing the small 72-image training set from overwriting general-purpose visual representations. The backbone produces a 7×7×2048 feature tensor. The task-specific head: Flatten (→100,352-dim) → Dropout(0.5) → Dense(128, ReLU) → Dropout(0.3) → Dense(1, Sigmoid). The sigmoid output $\sigma(z)\in[0,1]$ represents Parkinson's class confidence:

$$\sigma(z) = 1 / (1 + e^{-z}) \quad (2)$$

Compiled with Adam (η=0.0001) and binary cross-entropy loss:

$$L = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})] \quad (3)$$

Training ran for 5 epochs, batch size 32. Table II summarises the configuration. Figure 2 shows the layered architecture.

TABLE II

ResNet50 Model Configuration

Parameter	Value
Backbone	ResNet50 (ImageNet, frozen)
Input Shape	224×224×3
Flatten Output	100,352 units
Dropout (1)	Rate = 0.5
Dense Hidden	128 units, ReLU
Dropout (2)	Rate = 0.3
Dense Output	1 unit, Sigmoid
Total Params	36,433,025
Trainable	12,845,313
Optimiser	Adam (lr=0.0001)
Loss	Binary Cross-Entropy
Epochs	5
Batch Size	32

IX. FLASK DEPLOYMENT FRAMEWORK

The trained model and Grad-CAM module are integrated into a Flask web application providing a browser-accessible interface. The core prediction endpoint is a POST route accepting multipart image uploads. Upon

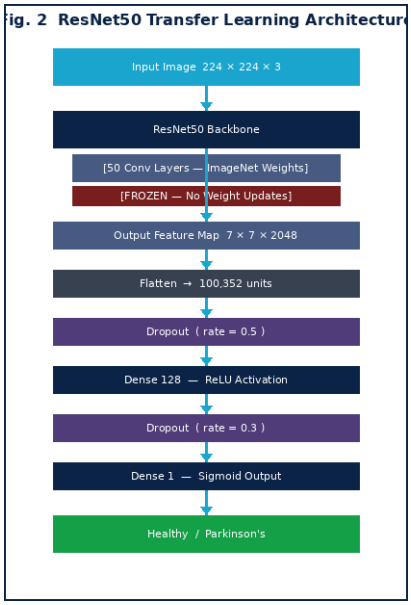


Fig. 2. Proposed ResNet50 transfer learning architecture. Frozen backbone extracts 7×7×2048 feature maps; trainable head produces binary PD prediction.

class score y^c with respect to feature maps A^k of the target convolutional layer is global-average-pooled to obtain importance weights α^c_k:

$$\alpha^c_k = (1/Z) \Sigma_i \Sigma_j \partial y^c / \partial A^k_{ij} \quad (4)$$

where Z is the number of spatial positions. The Grad-CAM heatmap L^c is the ReLU-gated weighted combination of feature maps:

$$L^c = \mathrm{ReLU}(\Sigma_k \alpha^c_k \cdot A^k) \quad (5)$$

Grad-CAM is computed with respect to conv5_block3_out—the terminal ResNet50 convolutional layer—producing feature maps at the highest semantic abstraction while retaining sufficient spatial resolution. The 7×7 heatmap is upsampled to 224×224, normalised to [0,1], and mapped to a colour jet (blue→yellow→red) via OpenCV cv2.applyColorMap(), then blended with the input spiral image. No retraining is required. Figure 5 shows representative Grad-CAM outputs.



Fig. 5. Grad-CAM heatmap overlays. (a) Healthy spiral: diffuse low-intensity blue activation. (b) Parkinson's spiral: concentrated warm red/yellow activations at tremor-affected stroke regions confirming clinically meaningful model attention.

receipt, the image is preprocessed as described in Section VI and passed to model.predict(). The scalar sigmoid output is compared to THRESHOLD=0.95; values ≥0.95 are assigned the Parkinson's label, otherwise Healthy. This conservative threshold minimises false positives—a priority in clinical screening contexts.

The response includes: (i) binary classification label; (ii) numeric confidence score as a percentage; (iii) Grad-CAM heatmap overlay as a base64-encoded PNG for inline display; and (iv) CSV-formatted prediction log capturing filename, label, confidence, and timestamp. Figure 6 illustrates the Flask dashboard.

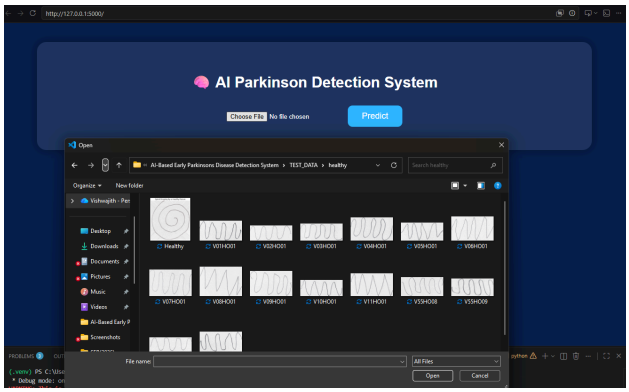


Fig. 6. Flask web application dashboard: image upload interface, classification result, sigmoid confidence score, Grad-CAM heatmap overlay, and prediction log table.

X. EXPERIMENTAL RESULTS AND DISCUSSION

The ResNet50 model was trained for five epochs on 72 spiral images and evaluated on a balanced 30-image test set (15 Healthy, 15 Parkinson's) in TensorFlow 2.x/Keras with Adam (lr=0.0001) and binary cross-entropy loss.

A. Training and Validation Curves

Figure 3 presents the training and validation accuracy and loss curves over five epochs. Training accuracy increases from ~0.44 to 0.75; validation accuracy peaks at 0.77 at epoch 1, stabilises, then reaches 0.80 at epoch 4. Both loss curves decrease monotonically (training: 3.8→0.57; validation: 1.65→0.47), demonstrating controlled convergence without divergence or overfitting. The modest train-validation gap at epoch 4 confirms effective Dropout regularisation.

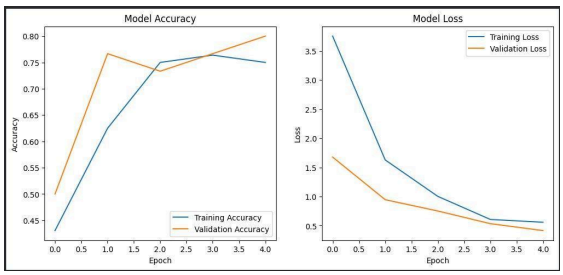


Fig. 3. Training/validation accuracy (left) and loss (right) over five epochs. Validation accuracy converges to 80%; both losses decrease monotonically without divergence.

B. Classification Report

Table III presents per-class precision, recall, F1-score and support values with macro-averaged metrics. The 80% accuracy is notable given only 72 training images, demonstrating compelling ResNet50 feature transfer effectiveness.

TABLE III

Classification Metrics — Test Set (n=30)

Class	Prec.	Recall	F1	Supp.
Healthy	0.76	0.87	0.81	15
Parkinson's	0.85	0.73	0.79	15
Macro Avg	**0.81**	**0.80**	**0.80**	**30**

For the Healthy class, recall (0.87) exceeds precision (0.76): 13 of 15 healthy subjects correctly classified. For Parkinson's, precision (0.85) exceeds recall (0.73): when the model assigns a Parkinson's label it is correct 85% of the time, but 4 of 15 true PD cases are missed. Clinically, false negatives carry greater risk—motivating future threshold calibration.

C. Confusion Matrix Analysis

Figure 4 displays the confusion matrix on the 30-image test set. Of 15 Healthy samples: 13 correctly classified (TN), 2 incorrectly labelled Parkinson's (FP). Of 15 Parkinson's samples: 11 correctly identified (TP), 4 missed (FN). Sensitivity=73.3%; Specificity=86.7%. The model is more reliable in ruling out PD than detecting it—consistent with the conservative THRESHOLD=0.95.

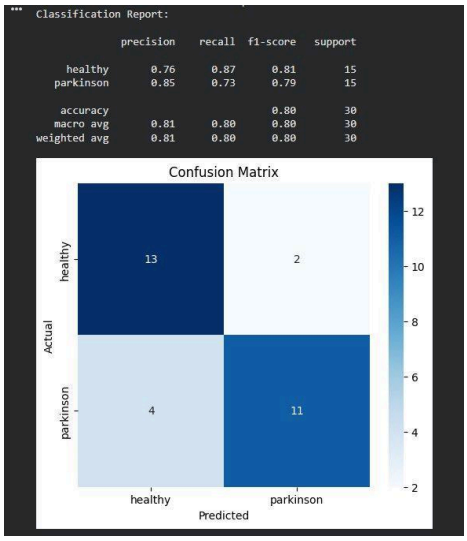


Fig. 4. Confusion matrix on 30-image test set. TP=11, TN=13, FP=2, FN=4. Sensitivity=73.3%; Specificity=86.7%.

D. Grad-CAM Visualisation Analysis

Figure 5 illustrates representative Grad-CAM outputs for both classes. For Healthy inputs, heatmaps show diffuse, cool blue colouring—no localised region dominates, consistent with smooth and geometrically regular drawing. For Parkinson's inputs, pronounced warm-colour (red/yellow) activation concentrates at peripheral strokes exhibiting tremor-induced deviations, aligning with the clinical understanding that PD motor dysfunction produces measurable irregularities at trajectory extremities.

XI. ADVANTAGES OF PROPOSED SYSTEM

- **Non-Invasive Screening:** Requires only a digital spiral drawing; eliminates invasive procedures, radioactive agents, or specialised instruments.
- **Data-Efficient Learning:** ResNet50 transfer learning achieves 80% accuracy using only 72 training images, demonstrating practical utility in low-data biomedical scenarios.
- **Clinician-Interpretable:** Grad-CAM heatmaps provide spatial justifications for each prediction without architectural modification or retraining.
- **End-to-End Deployment:** Flask delivers a complete pipeline from image submission to labelled prediction, heatmap visualisation, and CSV logging in a single browser session.
- **Regulatory Alignment:** Structured logging and XAI-generated justifications support audit and transparency requirements in medical AI governance.
- **Scalability:** Modular architecture supports extension to additional neurological tasks, backbone alternatives, and expanded datasets without fundamental redesign.

XII. LIMITATIONS

Several limitations constrain clinical applicability. First, the 72-image training corpus is substantially smaller than robust medical classifier development requires; heterogeneous clinical population performance has not been validated. Second, THRESHOLD=0.95 was empirically selected without formal calibration on a dedicated validation set. Third, Grad-CAM provides spatial attention but not feature-level explanations or quantitative uncertainty estimates. Fourth, the Flask application lacks user authentication, encrypted storage, or EHR integration—prerequisites for hospital deployment.

XIII. FUTURE SCOPE

Expanding the training corpus using PPMI or PaHaW combined with augmentation (rotation, elastic deformation, contrast adjustment) would improve robustness. Partial unfreezing of the ResNet50 backbone for domain-specific fine-tuning represents a promising accuracy improvement avenue. Replacing or augmenting the backbone with vision transformers (ViT) or EfficientNet may yield better feature extraction with fewer parameters. TensorFlow Lite porting would enable point-of-care mobile screening; HL7 FHIR API integration would link predictions to EHR records. Extension to voice tremor analysis (jitter, shimmer) and wearable accelerometer data would enable multi-modal PD screening.

XIV. CONCLUSION

This paper presented an AI-based Parkinson's Disease early detection system integrating ResNet50 transfer learning, Grad-CAM explainability, and Flask-based web deployment into a cohesive clinical screening pipeline. The system classifies spiral handwriting images with 80% accuracy on a balanced 30-image test set despite training on only 72 samples—demonstrating the power of transfer learning for biomedical image classification under data-scarce conditions.

The Grad-CAM module produces spatially meaningful heatmaps whose activation patterns align with clinically expected motor dysfunction signatures. The Flask web application realises an end-to-end deployment including inference, confidence scoring, heatmap visualisation, and prediction logging. The system is a proof-of-concept for non-invasive, explainable, and accessible AI-assisted neurological screening.

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